Assignment-01

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Sec: C

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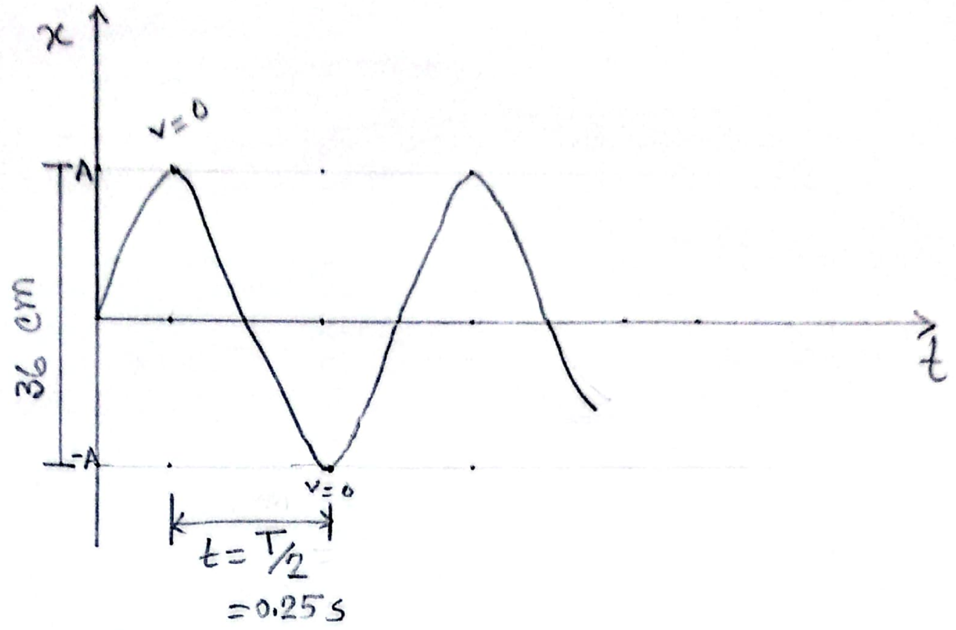

i) $\frac{T}{2} = 0.25\,s$ [from the figure]

$\therefore T = 0.5\,s$

$\therefore$ period 0.5 second

ii) frequency, $f = \frac{1}{T}$

$= \frac{1}{0.5}$ [from (i): $T = 0.5\,s$]

$= 2\,Hz$

iii) $2A = 36\,cm$ [from the figure]

$\therefore A = 18\,cm$

$\therefore$ amplitude of the motion is 18 cm.

2 ~~ma~~Given, mass, $m = 680\ g$

$= 0.68\ kg$

spring constant, $k = 65\ N/m$

amplitude, $A = 11\ cm$

$= 0.11\ m$

The oscillation of the block generate cosine form.

i) time period, $T = 2\pi\sqrt{\dfrac{m}{k}}$

$$= 2\pi\sqrt{\frac{0.68}{65}}$$

$$= 0.64\ s$$

ii) angular frequency, $\omega = \sqrt{\dfrac{k}{m}}$

$$= \sqrt{\frac{65}{0.68}}$$

$$= 9.78\ rad/s$$

iii) Phase constant, $\phi = 0°$

iv) $x = 0.11 \cos(9.78t)$

$$\Rightarrow v = \frac{dx}{dt} = -(0.11 \times 9.78) \sin(9.78t)$$

R

$$\therefore V_{(4sec)} = -(0.11 \times 9.78) \sin(9.78 \times 4)$$

$$= \boxed{-0.68 \text{ m/s}} \quad \times$$

$$\therefore V_{max} = -A\omega$$

$$= -(0.11 \times 9.78) = -1.0758 \text{ m/s}$$

v) $v = -(0.11 \times 9.78) \sin(9.78t)$

$$\Rightarrow a = \frac{dv}{dt} = -(0.11 \times 9.78^2) \cos(\cancel{\sin}\, 9.78t)$$

R

$$\therefore a_{(4 sec)} = -(0.11 \times 9.78^2) \cos(9.78 \times 4)$$

$$= \boxed{-8.16 \text{ m/s}^2} \quad \times$$

$$\therefore a_{max} = -10.52 \text{ m/s}^2$$

vi) $x = 0.11 \cos(9.78t)$ [from (iv)]

R

$$\therefore x_{(0 sec)} = 0.11 \cos(9.78 \times 0)$$

$$= 0.11 \text{ m}$$

$$\therefore x_{(7 sec)} = 0.11 \cos(9.78 \times 7)$$

$$= 0.087 \text{ m}$$

vii) Given, $A = 0.11$ m

from (ii), $\omega = 9.78$ rad/s

$\therefore v = \omega\sqrt{A^2 - x^2}$

$\therefore v_{(x=0.11)} = 9.78\sqrt{0.11^2 - 0.11^2}$

$= 0$

$\therefore v_{(x=0.04)} = 9.78\sqrt{0.11^2 - 0.04^2}$

$= 1.002$ m/s

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(3) Given,

mass, $m = 0.12$ kg

amplitude, $A = ~~0.~~ 8.5$ cm

$= 0.085$ m

time period, $T = 0.2$ s

i) $F_{max} = m a_{max}$

$= m A \omega^2 \quad [a(max) = A\omega^2]$

$= m A \left(\frac{2\pi}{T}\right)^2 \quad [\because \omega = \frac{2\pi}{T}]$

$= 0.12 \times 0.085 \times \left(\frac{2\pi}{0.2}\right)^2 = 10.067$ N

ii) $T = 2\pi\sqrt{\frac{m}{k}}$

$\Rightarrow 0.2 = 2\pi\sqrt{\frac{0.12}{k}}$

$\therefore k = 118.435 \ N/m$

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4) Given,

amplitude, $2A = 2 \ mm$

$\therefore A = 1 mm = 0.001 \ m$

frequency, $f = 120 \ Hz$

i) $A = 0.001 \ m$

ii) $V_{max} = A\omega$

$= A \times 2\pi f \quad [\omega = 2\pi f]$

$= 0.001 \times 2 \times \pi \times 120$

$= 0.754 \ m/s$

iii) $a_{max} = A\omega^2$

$= A \times (2\pi f)^2$

$= 0.001 \times (2\pi \times 120)^2$

$= 568.5 \ m/s^2$

5 Given,

$K = 400\ N/m$.

at time t,

$x = 0.1\ m$

$v = -13.6\ m/s$

$a = -123\ m/s^2$

i) $F = -Kx = ma$

$\Rightarrow \frac{K}{m} = -\frac{a}{x}$

$\Rightarrow \omega^2 = -\frac{a}{x}$ $\left[\because \omega = \sqrt{\frac{K}{m}}\right]$

$\Rightarrow \omega^2 = -\frac{(-123)}{0.1}$

$\Rightarrow 2\pi f = \sqrt{1230}$

$\therefore f = \frac{\sqrt{1230}}{2\pi}$

$= 5.58\ Hz.$

ii) $-kx = ma$ [from (i)]

$\therefore m = -\frac{kx}{a}$

$= -\frac{400 \times 0.1}{-123} = 0.325\ kg$

iii) $\omega = \sqrt{1230}$ rad/s [from (i)]

Now,

$$v = \omega\sqrt{A^2 - x^2}$$

$$\therefore A = \sqrt{\frac{v^2}{\omega^2} + x^2}$$

$$= \sqrt{\frac{(-13.6)^2}{1230} + (0.1)^2}$$

$$= 0.4005 \text{ m}$$

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6) mass, $m = 25 g = 0.025$ kg

spring constant, $k = 400$ dyns/cm

$= 0.4$ N/m

amplitude, $A = 10$ cm $= 0.1$ m

i) $$\omega = \frac{2\pi}{T} = \sqrt{\frac{k}{m}}$$

$$\Rightarrow T = 2\pi\sqrt{\frac{m}{k}}$$

$$= 2\pi\sqrt{\frac{0.025}{0.4}} = \boxed{1.57 \text{ s}}$$

ii) $\omega = 2\pi f = \sqrt{\frac{k}{m}}$

$\Rightarrow 2\pi f = \sqrt{\frac{0.025}{0.4}}$

$\therefore$ ~~f = 0.04 Hz.~~ 0.636

iii) $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{0.025}{0.4}} = \cancel{0.25}$ 4 rad/s

iv) $V_{max} = -A\omega$

$= -(0.1 \times \cancel{0.25}\ 4)$

~~$= -0.025$~~ $= -0.4\ ms^{-1}$

7 mass, $m = 1.68 \times 10^{-27}$ kg

frequency, $f = 10^{4}$ Hz

amplitud, $A = 10^{-10}$ m

$F = kx$

$= \omega^2 m x$

$= 4\pi^2 f^2 m A \quad [\because x = A]$

$= 4\pi^2 \times (10^4)^2 \times 1.68 \times 10^{-27} \times 10^{-10}$

$= 6.63 \times 10^{-8}$ N

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8. at mean position, $\boxed{v = v_{max} = 1\ m/s}$

at extremity position, $a = a_{max} = 1.57\ m/s^2$

Now,

$\boxed{v = A\omega} = 1$

$\boxed{a = A\omega^2} = A\omega \cdot \omega = 1 \times \omega$

$\Rightarrow \boxed{\omega = a} = 1.57$

$\Rightarrow \frac{2\pi}{T} = 1.57$

$\therefore T = \frac{2\pi}{1.57} = 4.002\ s$

9. amplitud, $A = 5\ m$, mean $\boxed{\text{distance}, x = 3\ m}$

at, $x = 3m$; $a = 48\ m/s^2$

i) $F = kx$

$\Rightarrow ma = kx$

$\Rightarrow \frac{k}{m} = \frac{a}{x}$

$\Rightarrow \omega^2 = \frac{a}{x}$

$\therefore \omega = \sqrt{\frac{a}{x}} = \sqrt{\frac{48}{3}} = 4\ rad/s$

$$\therefore v = \omega\sqrt{A^2 - x^2}$$
$$= 4\sqrt{5^2 - 3^2}$$
$$= 16 \text{ m/s}$$

ii) $\omega = \frac{2\pi}{T}$

$$\therefore T = \frac{2\pi}{\omega} = \frac{2\pi}{4} = 1.57 \text{ s}$$

iii) $\boxed{v_{max} = A\omega = 5 \times 4 = 20 \text{ m/s}}$

(10) frequency, $f = 0.25$ Hz

at, $t = 0$; $x = 0.37$ cm and $v = 0$ m/s

i) $T = \frac{1}{f} = \frac{1}{0.25} = 4$ sec

ii) $\omega = \frac{2\pi}{T} = \frac{2\pi}{4} = 1.57$ rad/s

iii) Here,

$x = 0.37 \text{ cm} = x_{max}$ [$\because v = 0$]

$\therefore A = 0.37 \text{ cm} = 0.0037 \text{ m}$

iv) $x = A\cos\omega t$

$\Rightarrow x = 0.0037\cos(1.57t)$

$\therefore x_{(t=3s)} = 0.0037\cos(1.57\times 3)$

$= -8.84\times 10^{-6}\ m$

v) $x = 0.0037\cos(1.57t)$

$\Rightarrow v = \frac{dx}{dt} = -(1.57\times 0.0037)\sin(1.57t)$

$\therefore v_{(t=3s)} = -(1.57\times 0.0037)\sin(1.57\times 3)$

$= 5.97\times 10^{-3}\ m/s$

(11) $x = 10\cos\left(3\pi t + \frac{\pi}{3}\right)$

i) $x_{(t=2.5s)} = 10\cos\left(3\pi\times 2.5 + \frac{\pi}{3}\right)$

$= 8.66\ m$

ii) $v = \frac{dx}{dt} = -(3\pi\times 10)\sin\left(3\pi t + \frac{\pi}{3}\right)$

$\therefore v_{(t=3s)} = -(3\pi\times 10)\sin\left(3\pi\times 3 + \frac{\pi}{3}\right)$

$= 81.62\ m/s$

iii) $a = \frac{dv}{dt} = -(30\pi \times 3\pi)\cos(3\pi t + \frac{\pi}{3})$

$\therefore a_{(t=2s)} = -(90\pi^2)\cos(3\pi \times 2 + \frac{\pi}{3})$

$= -444.13 \ m/s^2$

iv) $v = -30\pi \sin(3\pi t + \frac{\pi}{3})$ [from (ii)]

$\therefore v_{max} = -30\pi = -94.25 \ m/s$

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12 $x = 10\sin(10t - \frac{\pi}{6})$

Here, $A = 10m$, $\omega = 10$ rad/s

i) $\omega = 2\pi f = 10$

$\therefore f = \frac{10}{2\pi} = 1.6$ Hz

ii) $\omega = \frac{2\pi}{T} = 10$

$\therefore T = \frac{2\pi}{10} = 0.628$ s

iii) $x_{max} = A = 10m$

$\therefore x_{max} = 10m.$

iv) $x = 10\sin(10t - \frac{\pi}{6})$

$\Rightarrow v = \frac{dx}{dt} = 100\cos(10t - \frac{\pi}{6})$

$\therefore v_{max} = 100$ m/s

v) $a = \frac{dv}{dt} = -1000\sin(10t - \frac{\pi}{6})$

$\therefore a_{max} = -1000$ m/s^2

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13 amplitude, $A = 4$ m

frequency, $f = 0.5$ Hz

phase angle, $\phi = \frac{\pi}{4}$

i) $T = \frac{1}{f} = \frac{1}{0.5} = 2$ s

ii)

$\omega = 2\pi f = 2\pi \times 0.5 = \pi$ rad/s

$\therefore x = 4\cos(\cancel{\omega}\pi t + \frac{\pi}{4})$

iii) $x = 4\cos(\pi t + \pi/4)$

$\Rightarrow v = \frac{dx}{dt} = -4\pi \sin(\pi t + \pi/4)$

$\Rightarrow a = \frac{dv}{dt} = -4\pi^2 \cos(\pi t + \pi/4)$

$\therefore v_{(t=5s)} = -4\pi \sin(5\pi + \pi/4)$

$= 8.886$ m/s

$\therefore a_{(t=5s)} = -4\pi^2 \cos(5\pi + \pi/4)$

$= 27.9$ m/s^2

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(14) mass, $m = 2$ kg

spring constant, $k = 196$ N/m

~~a~~ amplitude, $A = 5$ cm $= 0.05$ m

i) $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{196}{2}} = 7\sqrt{2} = 9.899$ rad/s

≈ 10 rad/s

Now,

$\omega = 2\pi f = 7\sqrt{2}$

$\therefore f = \frac{7\sqrt{2}}{2\pi} = 1.576$ Hz

again,

$\omega = \frac{2\pi}{T} = 7\sqrt{2} \Rightarrow T = \frac{2\pi}{7\sqrt{2}} = 0.635$ s

ii) $\omega = 10$ rad/s

$A = 0.05$ m

$\therefore x = 0.05 \cos(10t)$

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mass, $m = 500$ gm $= 0.5$ kg

at $x = 3$ cm, $v = 40$ cm/s $= 0.4$ m/s

i) $F = k \cdot x$

$\Rightarrow mg = kx$

$$\therefore \boxed{k} = \frac{mg}{x} = \frac{0.5 \times 9.8}{7 \times 10^{-2}} = 70 \text{ N/m}$$

ii) $$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{70}{0.5}} = 11.83$$

iii) $$\omega = \frac{2\pi}{T} = \sqrt{\frac{k}{m}}$$

$$\therefore T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{0.5}{70}} = 0.53 \text{ s}$$

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mass, $\boxed{m = 25\ g = 0.025\ kg}$

spring constant, $k = 400$ dynes/cm

$$= \frac{400 \times 10^{-5}}{10^{-2}} \ N/m$$

$$= 0.4 \ N/m$$

$x = 10\ cm = 0.1\ m$

imparting velocity, $v = 40\ cm/s = 0.4\ m/s$

a) $T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{0.025}{0.4}} = 1.57\ s$

b) $f = \frac{1}{T} = \frac{1}{1.57} = 0.637\ Hz$

c) $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{0.4}{0.025}} = 4\ rad/s$

d) $E = \frac{1}{2} \times 0.4 \times 0.1^2 + \frac{1}{2} \times 0.025 \times 0.4^2$

$= 0.004\ J$

e) $E = \frac{1}{2}kA^2 \Rightarrow A = \sqrt{\frac{2E}{k}} = \sqrt{\frac{2 \times 0.004}{0.4}} = 0.14\ m$

f) $x = A\cos\theta_0 \quad [\theta_0 = \omega t + \phi]$

$\Rightarrow 0.1 = 0.14\cos\theta_0$

$\therefore \theta_0 = \cancel{0.78}^\circ\ 44.4^\circ$

g) $E = \frac{1}{2} m V_{max}^2$

$\Rightarrow V_{max} = \sqrt{\frac{2E}{m}} = \sqrt{\frac{2\times 0.004}{0.025}} = 0.57\ m/s$

$\therefore V_{max} = -0.57\ m/s$

h) $a_{max} = -A\omega^2 = -0.14\times 4^2 = -2.24\ m/s^2$

i) $A = 0.14m,\ \omega = 4\ rad/s,\ \phi$

from (f),

$\theta_0 = 44.4^\circ = \omega t + \phi$

$\therefore \phi = 44.4^\circ\ [\because t = 0]$

$\therefore x = 0.14\cos(4t + 44.4^\circ)$

$v = -0.56\sin(4t + 44.4^\circ)$

$a = -2.24\cos(4t + 44.4^\circ)$

j) $x_{(t=\pi/8\,s)} = 0.14\cos\left(4\times\frac{\pi}{8} + 44.4^\circ\times\frac{\pi}{180^\circ}\right)$

$= -0.098\ m$

$v_{(t=\pi/8\,s)} = -0.56\sin\left(4\times\frac{\pi}{8} + 44.4^\circ\times\frac{\pi}{180^\circ}\right)$

$= -0.4\ m/s$

$a_{(t=\pi/8\,s)} = -2.24\cos\left(4\times\frac{\pi}{8} + 44.4^\circ\times\frac{\pi}{180^\circ}\right)$

$= 1.57\ m/s^2$

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17. mass, $m = 2.72\times10^5\ kg$

frequency, $f = 10\ Hz$

amplitude, $A = 20\ cm = 0.2\ m$

i) $E = \frac{1}{2}kA^2$

$= \frac{1}{2}\omega^2 m A^2\ \left[\omega^2 = \frac{k}{m}\right]$

$= \frac{1}{2}4\pi^2 f^2 m A^2\ [\omega = 2\pi f]$

$= 2\pi^2\times10^2\times2.72\times10^5\times(0.2)^2$

$= 2.15\times10^7\ J$

ii) at equilibrium point, $v = v_{max}$

$$\therefore E = \frac{1}{2} m v_{max}^2$$

$$\therefore v_{max} = \sqrt{\frac{2E}{m}} = \sqrt{\frac{2 \times 2.15 \times 10^7}{2.72 \times 10^5}}$$

$$= 12.57 \text{ m/s}$$

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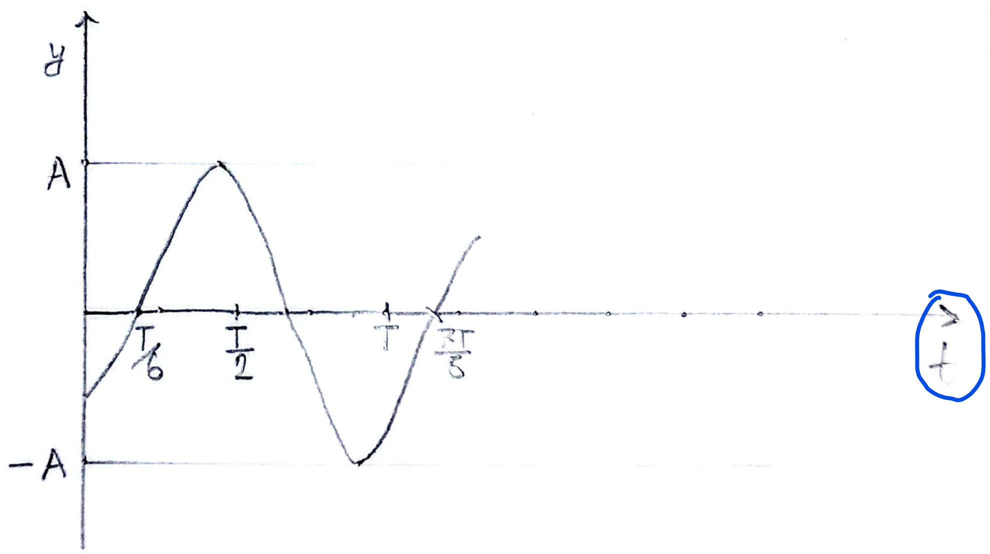
(18)

(i) $y = A \sin\left(\omega t - \frac{\pi}{3}\right)$

$$\omega t - \frac{\pi}{3} = 0$$

$$\frac{2\pi t}{T} = \frac{\pi}{3}$$

$$\therefore t = \frac{T}{6}$$



displacement (y) vs time (t) graph.

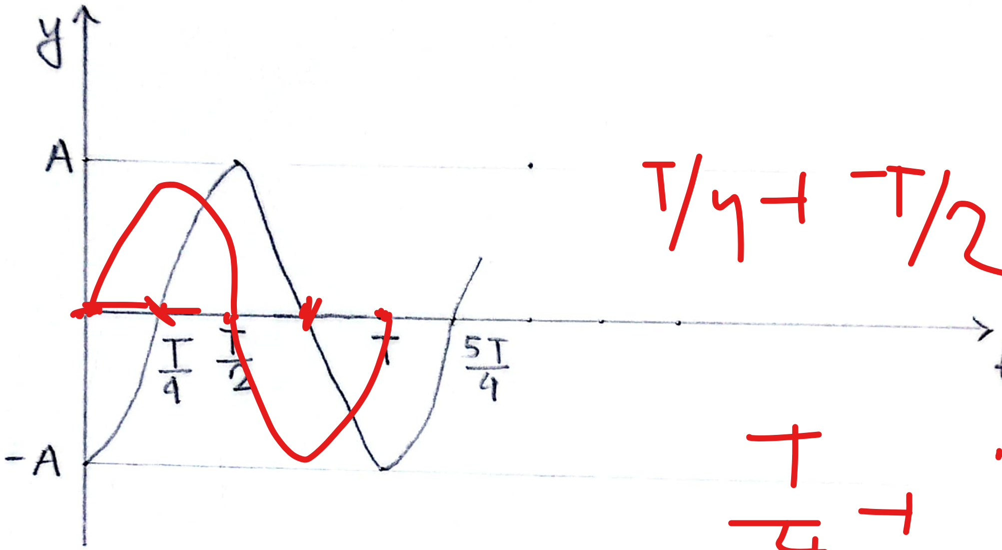
(ii) $y = A\sin\left(\omega t - \frac{\pi}{2}\right)$

$\cancel{\Rightarrow y = -A\cos(\omega t)}$

$\omega t - \frac{\pi}{2} = 0$

$\frac{2\pi t}{T} = \frac{\pi}{2}$

$\therefore t = \boxed{\frac{T}{4}}$



$T/4 + T/2 = \frac{3T}{4}$

$\frac{T}{4} + \frac{T}{2} +$

~~displacement (y) vs time (t) graph.~~

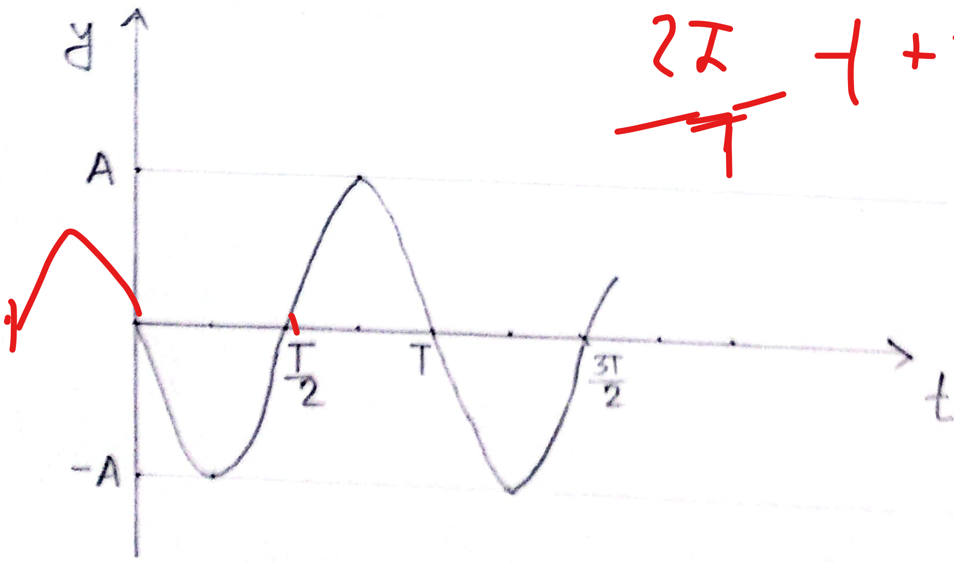
(iii) $y = A\sin(\omega t + \pi)$

$\frac{2\pi}{T} t + \pi = 0$

$\omega t + \pi = 0$

$\frac{2\pi t}{T} = -\pi$

$\therefore t = -\frac{T}{2}$



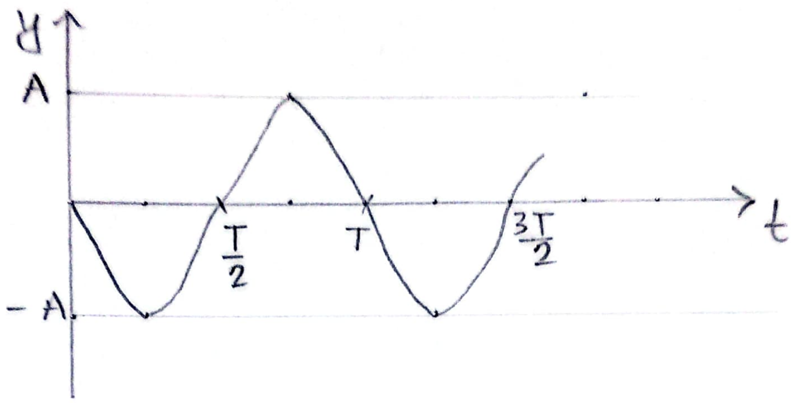
displacement (y) vs time (t) graph

(iv) $y = A\sin(\omega t - \pi)$

$\omega t - \pi = 0$

$\frac{2\pi t}{T} = \pi$

$t = \frac{T}{2}$



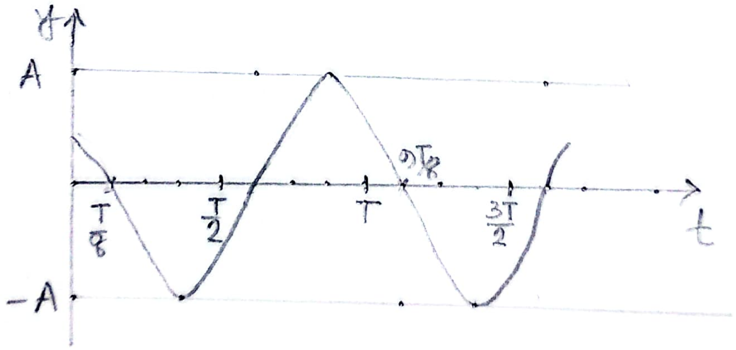
displacement (y) vs time (t) graph.

(v) $y = A\sin\left(\omega t + \frac{3\pi}{4}\right)$

$\omega t = -\frac{3\pi}{4}$

$\frac{2\pi t}{T} = -\frac{3\pi}{4}$

$\therefore t = \boxed{-\frac{3T}{8}}$



displacement (y) vs time (t) graph